"I Knew How to Report, But Didn't": Awareness, Helplessness, and Platform Distrust in CSAM Reporting Among University Students

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***Abstract*—This paper presents a cross-sectional survey study ($n = 47$) examining how university students at the University of Waterloo perceive, respond to, and reason about child sexual abuse material (CSAM) encountered on social media platforms. Data were collected between March and April 2024. Among respondents who confirmed an encounter ($n = 25$, 53.2%), 80% reported knowing how to file a report, yet only 44% did so. Of those who reported, only one participant felt the platform took appropriate action; nine received no response or no visible action, and one described the outcome as "never heard back." These findings reveal a structural reporting gap driven by two compounding dynamics: a bystander effect shaped by platform scale and perceived diffusion of responsibility, and platform distrust arising from repeated experiences of ineffectual moderation. Helplessness was the second most common emotional response (48%), and 32% of those who encountered CSAM ignored the content entirely. Against a backdrop of 36.2 million National Center for Missing and Exploited Children (NCMEC) CyberTipline reports and over 40,000 Cybertip.ca reports in Canada during 2023, and consistent user-side evidence that reporting mechanisms are neither trusted nor effective, this paper argues that the problem is not one of digital literacy but of structural accountability. Platform response quality, interface design, and mandatory outcome transparency are identified as the most likely levers for closing the gap between knowing how to report and actually doing so.**

***Index Terms*—child sexual abuse material, CSAM, social media, online reporting, bystander effect, platform accountability, digital safety, content moderation, user behaviour, university students, Canada**

I. Introduction

The proliferation of Child Sexual Abuse Material (CSAM) on mainstream social media platforms represents one of the most urgent and persistent failures of digital governance. In 2023, the National Center for Missing and Exploited Children (National Center for Missing and Exploited Children (NCMEC)) received 36.2 million reports of suspected child sexual exploitation via its CyberTipline, encompassing more than 104 million files [1]. That figure represents a 12% year-over-year increase and a growth of approximately 75% over five years [2]. In Canada, Cybertip.ca, the national tipline operated by the Canadian Centre for Child Protection (Canadian Centre for Child Protection (C3P)), recorded a 45% increase in online sexual exploitation reports between 2022 and 2023, processing over 40,000 public reports in that year [3]. Online sexual luring reports received by Cybertip.ca rose by more than 815% between 2018 and 2022 [4].

These institutional figures capture only the volume of content reported to designated authorities. They do not capture what happens in the seconds after an ordinary user scrolls past CSAM on a mainstream platform: whether they report it, ignore it, feel capable of doing something, or feel that doing something matters. That user-side dimension, the gap between encounter and action, is the focus of this paper.

The gap sits at the intersection of several well-documented phenomena. The online bystander effect, an extension of Latané and Darley's [5] classic model of diffusion of responsibility, predicts that in large, anonymous online environments, individuals are less likely to intervene because they assume others will [6], [7]. Platform interface design has been shown to actively discourage reporting through friction, opaque outcomes, and disenchantment with reactive moderation [8], [9]. Prior research on harmful content encounters documents a pattern of emotional helplessness that, when paired with perceived platform inefficacy, produces passive disengagement as the default response [7], [10].

Despite the severity of the problem, empirical data on how ordinary users, particularly those in higher education, experience and respond to CSAM on mainstream platforms remain scarce. Existing literature focuses primarily on law enforcement challenges [11], platform-side detection technology [1], and regulatory frameworks [12], [13], leaving the user-side behavioural and perceptual dimensions understudied. This paper addresses that gap through a cross-sectional survey of 47 University of Waterloo students.

The study addresses three research questions. **RQ1:** How

prevalent is CSAM exposure among university students, and on which platforms does it occur? **RQ2:** What emotional responses and behavioural choices characterise the immediate aftermath of a CSAM encounter, and what explains the gap between knowing how to report and actually doing so? **RQ3:** How do students assess current platform accountability, and what interventions do they prefer?

The dataset underlying this study is provided as a supplementary file accompanying this paper, with all responses anonymised in their original collection format.

II. Background and Related Work

A. The Scale of CSAM on Social Media

The growth of CSAM on social media platforms is documented primarily through institutional reporting. NCMEC's CyberTipline received 36.2 million reports in 2023, compared to 18.4 million in 2018 [1]. A 2023 Stanford Internet Observatory investigation found a network of more than 500 accounts on Instagram advertising self-generated CSAM with tens of thousands of likely buyers [14]. The investigation revealed safety failures in Instagram's own reporting interface for accounts with evident child exploitation indicators. In January 2024, the U.S. Senate Judiciary Committee convened a full hearing in which the CEOs of Meta, X, TikTok, Snapchat, and Discord testified about platform responses to child safety, with senators characterising the responses as inadequate and vague [12].

In Canada, the problem is equally acute. Statistics Canada's analysis of police-reported incidents found that in 2023, 87% of online child sexual exploitation incidents involved making or distributing child pornography [15]. C3P's Project Arachnid, an automated web crawler, processed over 164 billion images by January 2024 and has issued millions of removal notices to electronic service providers globally [15]. Despite this technical infrastructure, only a fraction of incidents come to the attention of police or dedicated tiplines, and research consistently shows that sexual offences against children are among the least reported of all crime categories [15].

Platform compliance with reporting obligations varies substantially. While Meta was responsible for more than 90% of all 2023 NCMEC CyberTipline reports, NCMEC noted that the majority of electronic service providers did not report at all, and many reports that were filed lacked sufficient detail for law enforcement action [1]. The Australian Institute of Criminology's analysis of platform CSAM responses found that detection technologies, including PhotoDNA hash-matching and AI classifiers, are effective against known content but poorly suited to new or self-generated material [11].

B. The Canadian Regulatory Context

Canada's regulatory response to online CSAM is shaped by two existing instruments and one landmark legislative attempt. The existing framework includes the *Criminal Code of Canada* provisions on child pornography (ss. 163.1) and *An Act Respecting the Mandatory Reporting of Internet Child Pornography by Persons Who Provide an Internet Service*, which requires internet service providers to report CSAM to C3P and to preserve associated data for law enforcement [13].

On February 26, 2024, the Government of Canada introduced Bill C-63, the proposed Online Harms Act. Had it been enacted, it would have imposed a 24-hour takedown obligation for CSAM and non-consensually distributed intimate images; established a Digital Safety Commission of Canada with enforcement powers; created a Digital Safety Ombudsperson to support users and victims; and required platforms to publish digital safety plans disclosing moderation statistics and internal research data [13]. Bill C-63 died on the order paper following the prorogation of Parliament in January 2025. At the time of this study's data collection (March–April 2024), the bill was before the House of Commons and represented the most substantive Canadian policy proposal in this space to that point.

The bill's 24-hour takedown obligation is directly relevant to the findings of this study. Users who reported CSAM and received no visible platform response would have had recourse to regulatory enforcement under the proposed framework. Its failure to pass means that Canadian users currently have no enforceable expectation of platform response, a structural condition that helps explain the low-trust, low-reporting equilibrium this study documents.

C. Bystander Theory and Online Harm Reporting

The bystander intervention model, first proposed by Latané and Darley [5] following the 1964 Kitty Genovese case, describes a five-stage cognitive process: noticing the event, interpreting it as an emergency, accepting personal responsibility, determining an appropriate response, and implementing that response. Failure at any stage produces non-intervention. The critical insight is that responsibility diffuses across larger audiences; the more bystanders are present, the less any individual feels obligated to act.

Online environments substantially complicate this model. The extreme scale of social media audiences amplifies diffusion of responsibility [7]. Anonymity reduces identifiability, further diminishing felt obligation [16]. The absence of non-verbal cues reduces the signal strength of distress and urgency [6]. Research on cyberbullying bystanders consistently finds that even when individuals perceive an event as harmful and know how to respond, many choose non-intervention, particularly when they doubt the efficacy of available response channels [9].

Self-efficacy regarding platform reporting tools is a significant predictor of whether bystanders act: those who believe reporting will produce an effect are more likely to report [7].

Prior victimisation or prior exposure to harmful content has a dual effect. Herry [6] found that individuals with prior experience of harassment were more likely to intervene on behalf of victims. However, when prior reporting experiences produce no visible outcome, the effect reverses: disenchantment with reporting processes reduces future reporting intention [8]. Wong et al. [8] specifically examined what drives use of reporting functions on social networking sites, finding that perceived reporting efficacy, rather than awareness of how to report, is the primary predictor of reporting behaviour. This is directly relevant to the gap this paper documents.

D. Platform Interface Design and Reporting Friction

Research on platform reporting interfaces has documented that their design systematically contributes to low reporting rates. Reporting flows are often opaque about what will happen after submission, do not communicate review timelines, provide no follow-up to reporters, and in several cases frame reports in ways that may deter low-confidence reporters [8], [9]. When users experience repeated encounters with moderation systems that produce no visible outcome, they engage in rational updating: the inference that reporting does not change anything [7].

This dynamic parallels the "privacy fatigue" construct documented in the digital privacy literature, in which repeated experiences of inadequate privacy protection lead users to disengage from privacy-protective behaviour entirely [17]. Applied to CSAM reporting, the equivalent phenomenon, which this paper terms reporting disenchantment, describes a state in which users retain reporting literacy but have learned to regard the reporting mechanism as ineffective and therefore do not deploy that literacy [8].

E. Altruism, Empathy, and the Motivational Basis of Reporting

The preceding subsections explain why individuals who encounter harmful content often do not report it. A complementary question, why some individuals do report, is equally important for understanding the full distribution of bystander responses.

Batson's empathy-altruism hypothesis [18] provides the foundational account. The hypothesis holds that empathic concern, an other-oriented emotional response to perceived need, produces genuinely altruistic motivation: a motivational state whose ultimate goal is increasing another's welfare rather than one's own. Critically, this motivation is distinct from egoistic alternatives such as aversive-arousal reduction or anticipated social approval. Batson's programme of experimental research has consistently found that when empathic concern is activated, individuals engage in helping behaviour even in conditions of easy escape and low social reward [18]. In the context of CSAM reporting, this framework predicts that reporters are not primarily motivated by compliance norms but by a genuinely other-oriented concern for the child depicted.

Empirical support for the empathy-reporting link comes from Vahed et al. [19], who found in an experimental study of social media users ($n = 294$) that trait empathy was positively associated with content reporting rates, while trust in social media platforms was negatively associated with reporting decisions. The latter finding is particularly significant for this paper: it suggests that the 56% of encountering respondents who did not report may have been suppressed not by indifference but by a pre-existing distrust of platform moderation systems, precisely the low-trust equilibrium that the reporting disenchantment framework predicts.

The emotional pathway from witnessing to action, however, is more differentiated than a simple empathy-to-helping chain. Loh and Galen [20] found that among cyberbullying bystanders, empathic anger, an other-oriented anger at the injustice done to a victim, was uniquely associated with defending intentions, whereas empathic distress, which focuses attention on the bystander's own discomfort, was associated with passive bystanding. This distinction has direct implications for this study's findings: the 68% of encountering respondents who reported feeling disgusted may be experiencing a blend of these two states, and whether disgust channels into outward-directed anger (motivating reporting) or inward-directed distress (motivating avoidance) may determine whether the bystander moves from emotional reaction to behavioural action.

The role of moral judgment as a mediator between empathy and helping behaviour further complicates this picture. Liu et al. [21] found that among adolescent cyberbullying bystanders, the empathy-to-intervention pathway was mediated by internet moral judgment: perceived severity of harm triggered moral evaluation, which in turn predicted helping behaviour. This chain suggests that even when empathy and disgust are high, the translation into action requires an intermediate moral determination that the situation demands response. Applied here, the near-universal seriousness rating in this study (mean 8.11/10) indicates that the moral determination, recognising the situation as requiring action, is broadly shared. The activation of altruistic motivation sufficient to overcome diffusion of responsibility and reporting disenchantment is what breaks down.

Finally, Scaffidi Abbate et al. [22] demonstrated in a controlled online experiment that prosocial priming, brief activation of altruism-related concepts in the encounter context, can partially offset the bystander effect even in virtual environments. This

finding is relevant to platform design: interfaces that frame reporting as an act of protection for a specific victim, rather than a generic content moderation action, may activate the altruistic motivation that diffusion of responsibility otherwise suppresses.

III. Methodology

A. Study Design

This study employed a cross-sectional online survey design. The survey examined five dimensions: (1) prior exposure to CSAM on social media; (2) platform distribution of encounters; (3) emotional reactions and behavioural responses upon encounter; (4) reporting knowledge, behaviour, and perceived platform responsiveness; and (5) policy preferences for platform accountability. The survey was administered via Google Forms between March 11 and April 10, 2024, and distributed through course networks at the University of Waterloo.

B. Participants

A total of 47 responses were collected. Participants ranged in age from 18 to 24 years (mean: 21.0). Gender distribution: male ($n = 21$, 44.7%), female ($n = 14$, 29.8%), non-binary ($n = 5$, 10.6%), not disclosed ($n = 5$, 10.6%), agender ($n = 1$), and queer ($n = 1$). Faculty representation: Mathematics ($n = 23$, 48.9%), Engineering ($n = 9$, 19.1%), Arts ($n = 10$, 21.3%), Science ($n = 4$, 8.5%), and one graduate student in Engineering. One additional response listed Accounting as faculty with a corresponding undergraduate term; this response was retained. Academic terms ranged from 1B to 4B with the majority in third and fourth year. The sampling frame is convenience-based and the sample is not representative of the University of Waterloo student population as a whole.

C. Survey Instrument

The survey comprised 15 items combining binary, multiple-choice, Likert-type, and open-response formats. Key items included: a binary yes/no/unsure exposure question; a multi-select platform identification question; a frequency rating (never/rarely/occasionally/frequently); an open-response content description; multi-select questions on personal and recommended reactions; a binary question on reporting knowledge; a follow-up on perceived platform response; an 11-point seriousness scale (0–10); a 5-point platform adequacy scale (1 = platforms do enough, 5 = platforms do not do enough); a 10-point reporting effectiveness scale; a multi-select policy preference question; and a yes/maybe/no item on desire for further education on CSAM reporting. The instrument defined CSAM inclusively as any involvement of minors in sexual contexts, covering visual media, online conversations, comments, posts, and related content.

D. Dataset Availability

The full anonymised dataset is provided as a supplementary file accompanying this paper. The dataset preserves all original response fields with no modification beyond the removal of the Google Forms submission timestamp (replaced with a date-only field) to prevent potential re-identification of responses submitted within narrow time windows.

E. Data Analysis

Descriptive statistics were computed for all quantitative variables. Subgroup comparisons were conducted between respondents who had and had not encountered CSAM on key outcome variables. Platform and action preference frequencies were tabulated. Open-response descriptions of encountered content were reviewed against the survey's CSAM definition. One open-response description ("a lot of physical violence in terms of hitting, a lot of pages post fights") did not meet the instrument's definition and was excluded from qualitative content analysis while retained in all quantitative item analyses.

F. Ethical Considerations

Participation was voluntary and anonymous. No images, videos, or detailed descriptions of abusive content were solicited. The survey directed participants to Cybertip.ca and the NCMEC CyberTipline for any content they wished to report during or after participation. The study was conducted in accordance with the Tri-Council Policy Statement on ethical research involving humans and did not require institutional ethics board review under the University of Waterloo's guidelines for low-risk anonymous survey research.

IV. Results

A. Exposure Prevalence and Platform Distribution

Of 47 respondents, 25 (53.2%) confirmed having encountered CSAM on social media, 21 (44.7%) had not, and one responded "maybe." Exposure frequency among those who confirmed an encounter: rarely/once or twice ($n = 13$, 52%); occasionally/a few times a month ($n = 7$, 28%); frequently/several times a month or more ($n = 7$, 28%).

Platform distribution among encountering respondents is shown in Table I. Instagram was the most frequently cited platform, appearing in 84% of encountering respondents' selections, followed by TikTok (56%), Twitter/X (48%), and YouTube (44%). Less frequently cited were Facebook, Snapchat, and Telegram. Several participants encountered CSAM across multiple platforms simultaneously.

B. Content Types Encountered

Where qualitative descriptions were provided, content spanned the full breadth of the survey's inclusive CSAM definition: explicit video and image depictions of minors being sexually

TABLE I
PLATFORM DISTRIBUTION OF CSAM ENCOUNTERS ($n = 25$; MULTIPLE PLATFORMS PER RESPONDENT PERMITTED)

Platform	Respondents	%
Instagram	21	84%
TikTok	14	56%
Twitter / X	12	48%
YouTube	11	44%
Facebook	8	32%
Snapchat	8	32%
Telegram	5	20%
Other	5	20%

assaulted; animated and illustrated sexual content depicting characters identified as minors, including anime and cartoon imagery; children in non-sexualized contexts accompanied by sexualizing comments; parent-managed accounts described as venues for exploitation; and derogatory or grooming-adjacent comments directed at minors. Several respondents declined to describe content and noted only the platform. The breadth of types reflects the survey's inclusive definition and is consistent with research indicating that user-side exposure to CSAM-adjacent content substantially exceeds exposure to content meeting strict legal definitions [11].

C. Emotional Reactions and Behavioural Responses

Reactions among the 25 confirming respondents are presented in Table II. Disgust was the most common response (68%), followed by helplessness (48%). Reporting to the platform occurred in 11 of 25 cases (44%), while 8 (32%) ignored the content entirely. Blocking the poster's account was selected by 8 respondents (32%). The co-occurrence of helplessness and non-reporting is a key pattern: of the 8 who ignored the content, 6 also indicated feeling helpless, suggesting that passive disengagement reflected perceived futility rather than indifference.

TABLE II
REPORTED REACTIONS AMONG RESPONDENTS WHO ENCOUNTERED CSAM ($n = 25$)

Reaction	n	%
Felt disgusted or disturbed	17	68%
Felt helpless	12	48%
Reported to the platform	11	44%
Blocked the account	8	32%
Ignored it	8	32%

D. The Reporting Gap: Knowledge, Behaviour, and Platform Response

The study's central finding is a 36-percentage-point structural gap between reporting knowledge and reporting behaviour. Among the 25 who encountered CSAM, 20 (80%) knew how to report, 5 (20%) were unsure, and none reported definitively not knowing. Despite this high rate of reporting literacy, only 11 of 25 (44%) actually filed a report.

Among the 11 who did report, perceived platform response: appropriate action taken ($n = 1$, 9%); no action taken ($n = 9$, 82%); unsure/no response ($n = 1$, 9%). One participant specifically described the platform response as "never heard back." Of the 25 who encountered CSAM, only one (4%) both filed a report and believed the platform acted appropriately.

E. Perceived Seriousness, Platform Adequacy, and Reporting Effectiveness

Across all 47 respondents, perceived seriousness of CSAM as an issue (0–10 scale) yielded a mean of 8.11 (SD = 2.04). Those who had encountered CSAM rated mean seriousness at 8.40 versus 7.71 for those who had not. Notably, even those who had never encountered CSAM rated its seriousness at nearly 8 out of 10, indicating high baseline awareness independent of direct exposure.

Platform adequacy (1 = platforms do enough, 5 = platforms do not do enough) yielded a full-sample mean of 2.40, placing collective opinion in the mid-range leaning toward "do not do enough." Reporting effectiveness (10-point scale) yielded a mean of 4.29 (SD = 2.07), indicating moderate to low perceived effectiveness. There was no meaningful difference in effectiveness ratings between those who had and had not used the reporting system, suggesting that low perceived effectiveness is a generalised population-level belief, not an outcome of direct reporting experience.

F. Policy Preferences

Results are presented in Table III. Content removal (85%) and account suspension from posting (85%) were the most widely supported interventions, followed by engagement disabling (72%). Data forfeiture to authorities was supported by 55%. A "warn users and allow them to proceed" option was selected by only 34%, indicating that most respondents do not support a permissive, notification-only intervention model.

Desire for further education on CSAM and how to report it was endorsed by 21 respondents (45%), with 14 (30%) unsure and 12 (26%) declining. The majority (75%) expressed some openness to additional education, relevant to this paper's argument that the problem is not ignorance of mechanisms but distrust of their efficacy.

TABLE III
PREFERRED PLATFORM RESPONSES TO CSAM ($n = 47$; MULTIPLE SELECTIONS PERMITTED)

Preferred Action	n	**%**
Take down content	40	85%
Disable account from posting	40	85%
Disable engagement with content	34	72%
Forfeit data to authorities	26	55%
Warn user and allow to proceed	16	34%

V. DISCUSSION

A. The Reporting Gap as a Structural Problem

The 36-percentage-point gap between knowing how to report (80%) and actually reporting (44%) resists explanation through digital literacy frameworks. Respondents who did not report were not ignorant of the reporting mechanism; they were uncertain whether using it would matter. This is supported by the finding that reporting effectiveness was rated at 4.29/10 across the full sample, including among respondents who had never used the reporting system. The perception of inefficacy precedes and shapes behaviour independently of direct experience.

This pattern is consistent with Wong et al.'s [8] finding that perceived reporting efficacy, rather than awareness of how to report, is the primary driver of reporting behaviour on social networking sites. A user who has formed a prior belief that platform moderation is unresponsive carries that belief into any future encounter, and it suppresses the likelihood of reporting even when the user possesses full procedural knowledge. The near-total absence of positive platform responses in this dataset, with only 1 of 11 reporters believing appropriate action was taken, provides rational grounds for exactly this belief.

The finding also extends bystander theory into the CSAM domain. Diffusion of responsibility is structurally amplified on platforms with hundreds of millions of users: any individual encountering CSAM can reasonably infer that many others have already seen it and that some proportion of them have already reported it. Without outcome feedback from the platform, there is no signal to interrupt that inference. The default assumption, that the system is handling it without one's personal involvement, is maintained even when the system is not [5], [7].

B. Helplessness as the Operational State of Non-Reporters

The prevalence of helplessness (48% of encountering respondents), the second most common emotional response after disgust, is consistent with a reporting ecology that produces what Seligman [10] described as learned helplessness: repeated exposure to an unresponsive environment generates a generalised expectation of futility that inhibits action even when action is technically available. In this study, the co-occurrence of helplessness with the "ignored it" response (6 of 8 who ignored content also reported feeling helpless) is consistent with that interpretation.

This is a meaningfully different problem from low reporting literacy, and it requires different interventions. Educational campaigns explaining how to file a Cybertip.ca report or a NCMEC CyberTip will not address the fundamental question of whether filing that report accomplishes anything. That 75% of respondents expressed openness to further education suggests a genuinely receptive population, but only if that education is paired with credible evidence that reporting produces outcomes [8].

The altruism literature adds a further dimension. The 44% who did report in this study are best understood through Batson's empathy-altruism framework [18]: their empathic concern for the child depicted was sufficient to produce altruistic motivation strong enough to override diffusion-of-responsibility and low efficacy expectations. The 48% who felt helpless, by contrast, may have experienced what Loh and Galen [20] describe as empathic distress, an inward-oriented response that focuses attention on the bystander's own discomfort rather than the victim's need, and which is associated with passive bystanding rather than intervention. Disgust, the modal reaction in this study (68%), is an ambivalent emotional state in this regard: it can channel into empathic anger that motivates action, or into aversive-arousal that motivates avoidance [20]. Platform interfaces that reframe reporting as a direct, named act of protection for a specific victim, rather than a generic flag-for-review action, may help tilt that ambivalence toward the altruistic pathway [22].

C. Canadian and International Policy Implications

The policy preference data reveal a population broadly aligned on a stringent accountability framework. Content takedown (85%) and account suspension (85%) are near-universally supported, engagement disabling (72%) commands strong majority consensus, and only 34% support a warn-and-allow approach. These preferences align poorly with current platform practice, in which the majority of electronic service providers do not file NCMEC reports at all [1], and none are currently required to notify reporters of outcomes.

In the Canadian context, the student preferences documented here map closely onto the core obligations proposed in Bill C-63. The bill's 24-hour takedown requirement for CSAM and its obligation to notify users who flagged content of the resulting action [13] directly address the two failures this study documents: slow or absent content removal, and the absence of

reporter feedback. The bill's failure to pass before Parliament's prorogation in January 2025 leaves Canadian users without an enforceable standard for platform response, in a country where Cybertip.ca saw a 45% increase in exploitation reports in the year preceding this survey [3].

The 55% support for data forfeiture to authorities reflects majority appetite for law enforcement cooperation, though one participant explicitly noted reservations about the word "forfeit" and the scope of authority it implies, a concern that echoes civil liberties critiques of both Bill C-63 and its U.S. counterparts such as the Eliminating Abusive and Rampant Neglect of Interactive Technologies (EARN IT) Act [12]. This ambivalence suggests that policy design will need to distinguish between mandatory NCMEC-style reporting, which commands strong support, and broader data access arrangements, which face more qualified endorsement.

D. Instagram and TikTok as Priority Platforms

Instagram's appearance in 84% of encountering respondents' selections is consistent with the Stanford SIO investigation [14], which identified Instagram as the primary platform for CSAM network formation. TikTok's appearance in 56% reflects the platform's algorithmic recommendation architecture, which has been criticised for surfacing exploitation-adjacent content to users with no prior search history in that space. The dominance of mainstream platforms in this dataset challenges framing of CSAM as a dark-web or fringe-platform problem. Telegram's presence (20%) reflects its documented use for CSAM distribution through private channels [23], but its minority position relative to Instagram and TikTok underscores that enforcement focus on encrypted fringe channels misses the mainstream exposure that ordinary users encounter.

C3P's 2023–24 reporting highlighted similar platform dominance patterns in the Canadian context, with social media platforms identified as the primary venues for online luring and exploitation, and Cybertip.ca issuing alerts specifically for Wizz, Omegle, and Discord during the same period as this survey's data collection [4].

E. Limitations

This study has several limitations. The sample is small ($n = 47$) and drawn from a single university, with a strong skew toward the Faculty of Mathematics (48.9%). Responses are self-reported and subject to social desirability bias, particularly for items relating to whether respondents filed a report. The survey period coincides with the beginning of the era of AI-generated CSAM, a category not explicitly addressed in the instrument. The survey's inclusive definition of CSAM may have broadened the exposure estimate relative to narrower legal definitions, though this is consistent with research indicating that user-side exposure to CSAM-adjacent content is substantially higher than exposure to legally-defined content [11]. The cross-sectional design does not permit causal inference: the association between helplessness and non-reporting could reflect pre-existing attitudes toward platform efficacy rather than responses to specific encounters.

VI. Conclusion

The central argument of this paper is that the gap between knowing how to report CSAM and actually doing so reflects a structural failure of platform accountability rather than a deficit of user knowledge. Among University of Waterloo students who encountered CSAM, 80% knew how to report and only 44% did. Of those who reported, one in eleven believed the platform took appropriate action. These findings, situated against 36.2 million NCMEC CyberTipline reports and over 40,000 Cybertip.ca reports in 2023, and against a Canadian legislative landscape in which Bill C-63's enforceable 24-hour takedown and reporter-notification obligations died on the order paper, indicate that the user-side dimension of CSAM governance is shaped primarily by distrust of reporting outcomes rather than by ignorance of reporting mechanisms.

Three implications follow. First, platform accountability measures, specifically mandatory outcome reporting to users who file reports, standardised response timelines, and public disclosure of moderation action rates, are necessary to rebuild the trust that would activate the reporting literacy that already exists. Second, regulatory frameworks must address not only whether platforms file reports but whether those reports are of sufficient quality for law enforcement action and whether platforms communicate outcomes to the users who initiated them. Third, educational campaigns about CSAM reporting must be paired with honest, evidence-grounded communication about current system limitations and what reporting can and cannot be expected to accomplish.

The 36-percentage-point reporting gap documented here is a proxy for the larger accountability deficit in platform governance: users know what should happen, want it to happen, and have the technical means to initiate it, but have learned, through repeated experience or population-level inference, that the system does not respond. Closing that gap requires changing the system.

List of Acronyms

C3P Canadian Centre for Child Protection: A Canadian non-profit organization that operates Cybertip.ca, Canada's national tipline for reporting online child sexual exploitation, and Project Arachnid, an automated web crawler that detects CSAM and issues

removal notices to electronic service providers worldwide 1, 2, 7

CSAM Child Sexual Abuse Material 1–7

NCMEC National Center for Missing and Exploited Children 1, 2, 4, 6, 7

References

[1] National Center for Missing and Exploited Children. (2024) 2024 CyberTipline Report. [Online]. Available: https://ncmec.org/CyberTiplineData

[2] U.S. Department of Homeland Security. (2024, Apr.) KNOW2PROTECT™ SOCIAL MEDIA TOOLKIT: Communication Guidance and Resources. [Online]. Available: https://dhs.gov/sites/default/files/2024-04/24_0416_k2p_social-media-toolkit.pdf

[3] CyberTip.ca. (2024) Reports from the public processed in 2023. Canadian Centre for Child Protection. [Online]. Available: https://cybertip.ca/en/about/reports-from-public-processed-2023/

[4] ——. (2023, Feb.) Reports of online sexual luring of canadian kids up 815% in five years. Canadian Centre for Child Protection. [Online]. Available: https://cyberaide.ca/en/campaigns-and-media/news-releases/2023/safer-internet-day-2023/

[5] B. Latané and J. M. Darley, *The Unresponsive Bystander: Why Doesn't He Help?* New York, NY: Appleton-Century-Crofts, 1970.

[6] E. Herry and K. L. Mulvey, "Gender-based cyberbullying: Understanding expected bystander behavior online," *Journal of Social Issues*, vol. 79, no. 4, pp. 1210–1230, May 2022. [Online]. Available: https://doi.org/10.1111/josi.12503

[7] S. Jeyagobi, S. Munusamy, M. R. Kamaluddin, A. R. A. Badayai, and J. Kumar, "Factors influencing negative cyber-bystander behavior: A systematic literature review," *Frontiers in Public Health*, vol. 10, Oct. 2022. [Online]. Available: https://doi.org/10.3389/fpubh.2022.965017

[8] R. Y. M. Wong, C. M. K. Cheung, and B. Xiao, "Combating online abuse: What drives people to use online reporting functions on social networking sites," in *Proceedings of the 49th Hawaii International Conference on System Sciences*, ser. HICSS '16. Kauai, HI, USA: IEEE Computer Society, Washington, DC, USA, Jan. 2016, pp. 1–10. [Online]. Available: https://dl.acm.org/doi/10.1109/HICSS.2016.58

[9] L. Huang, W. Li, Z. Xu, H. Sun, D. Ai, Y. Hu, S. Wang, and Y. Li, "The severity of cyberbullying affects bystander intervention among college students: The roles of feelings of responsibility and empathy," *Psychology Research and Behavior Management*, vol. 16, pp. 893–903, Mar. 2023. [Online]. Available: https://doi.org/10.2147/PRBM.S397770

[10] M. E. P. Seligman, *Helplessness: On Depression, Development, and Death*. San Francisco, CA: W. H. Freeman and Company, 1975.

[11] C. Teunissen and S. Napier, "Child sexual abuse material and end-to-end encryption on social media platforms: An overview," in *Trends & Issues in Crime and Criminal Justice*. Australian Institute of Criminology, Australian Government, Jul. 2022, no. 653. [Online]. Available: https://aic.gov.au/sites/default/files/2022-07/ti653_csam_and_end-to-end_encryption_on_social_media_platforms.pdf

[12] U.S. Senate, Committee on the Judiciary, "Full Committee Hearing: Protecting Our Children Online," *U.S. Government Publishing Office*, no. S. Hrg. 118-028, Feb. 2023, 118th Congress, 1st Sssion, Serial No. J–118–3. [Online]. Available: https://judiciary.senate.gov/committee-activity/hearings/protecting-our-children-online

[13] House of Commons, Canada, "Bill C-63: An act to enact the online harms act, to amend the criminal code, the canadian human rights act and an act respecting the mandatory reporting of internet child pornography by persons who provide an internet service," *Parliament of Canada*, Feb. 2024, 44th Parliament, 1st Session, First Reading. [Online]. Available: https://parl.ca/DocumentViewer/en/44-1/bill/C-63/first-reading

[14] DiResta, Reneé, and Thiel, David, "An investigation into self-generated child sexual abuse material networks on social media," *Stanford Law School*, Jun. 2023, Original report at https://web.archive.org/web/20250816105037/https://cyber.fsi.stanford.edu/news/addressing-distribution-illicit-sexual-content-minors-online. [Online]. Available: https://law.stanford.edu/podcast/an-investigation-into-self-generated-child-sexual-abuse-material-networks-on-social-media/

[15] L. Savage, "Online child sexual exploitation: A statistical profile of police-reported incidents in Canada, 2014 to 2022," *Juristat, Statistics Canada, Government of Canada*, vol. 44, no. 1, Mar. 2024. [Online]. Available: https://www150.statcan.gc.ca/n1/pub/85-002-x/2024001/article/00003-eng.htm

[16] T. K. H. Chan, C. M. K. Cheung, I. Benbasat, B. Xiao, and Z. W. Y. Lee, "Bystanders join in cyberbullying on social networking sites: The deindividuation and moral disengagement perspectives," *Information Systems Research*, vol. 34, no. 3, pp. 828–846, Aug. 2022. [Online]. Available: https://doi.org/10.1287/isre.2022.1161

[17] X. Tian, L. Chen, and X. Zhang, "The role of privacy fatigue in privacy paradox: A PSM and heterogeneity analysis," *Applied Sciences*, vol. 12, no. 19, Sep. 2022. [Online]. Available: https://doi.org/10.3390/app12199702

[18] C. D. Batson, *Altruism in Humans*. New York, NY: Oxford University Press, 2011. [Online]. Available: https://doi.org/10.1093/acprof:oso/9780195341065.001.0001

[19] S. Vahed, C. Goanta, P. Ortolani, and A. G. Sanfey, "Moral judgment of objectionable online content: Reporting decisions and punishment preferences on social media," *PLOS ONE*, vol. 19, no. 3, Mar. 2024. [Online]. Available: https://doi.org/10.1371/journal.pone.0300960

[20] H. R. Steinvik, A. L. Duffy, and M. J. Zimmer-Gembeck, "Bystanders' responses to witnessing cyberbullying: the role of empathic distress, empathic anger, and compassion," *International Journal of Bullying Prevention*, vol. 6, pp. 399–410, Mar. 2023. [Online]. Available: https://doi.org/10.1007/s42380-023-00164-y

[21] Y. Hu, T. Zhang, H.-f. Shi, and C.-y. Fan, "Empathy and bystander helping behavior in cyberbullying among adolescents: the mediating role of internet moral judgment and the moderating role of internet self-efficacy," *Frontiers in Psychology*, vol. 14, Sep. 2023. [Online]. Available: https://doi.org/10.3389/fpsyg.2023.1196571

[22] C. Scaffidi Abbate, R. Misuraca, G. Vaccaro, M. Roccella, L. Vetri, and S. Miceli, "Prosocial priming and bystander effect in an online context," *Frontiers in Psychology*, vol. 13, Aug. 2022. [Online]. Available: https://doi.org/10.3389/fpsyg.2022.945630

[23] P. Bleakley, E. Martellozzo, R. Spence, and J. DeMarco, "Moderating online child sexual abuse material (CSAM): Does self-regulation work, or is greater state regulation needed?" *European Journal of Criminology*, vol. 21, no. 2, pp. 231–250, 2023. [Online]. Available: https://doi.org/10.1177/14773708231181361